PCEOT402
ELECTRICAL MACHINES

Module 2

Transformers

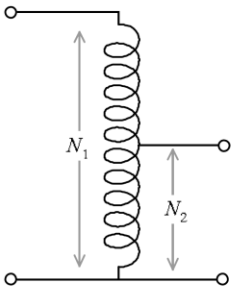


Transformer

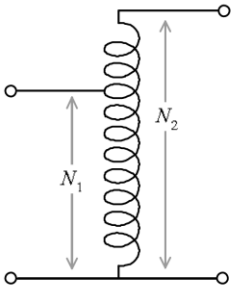
- Autotransformer – saving of copper
- 3- phase transformer – construction
- Different connections of 3-phase transformers
- Y-Y, Δ-Δ, Y-Δ, Δ-Y

Single Phase Autotransformers

Step down



Step up



- In autotransformers, a part of the winding is common for both primary and secondary
- Both step-down and step-up operation are possible
- Ratio of voltage and currents is similar to the two winding transformer (related to N_1 and N_2)
- Size and cost will be less compare to a two winding transformer of similar capacity.
- Primary and secondary windings are not electrically isolated
- A two winding transformer can be converted to an autotransformer by proper connections

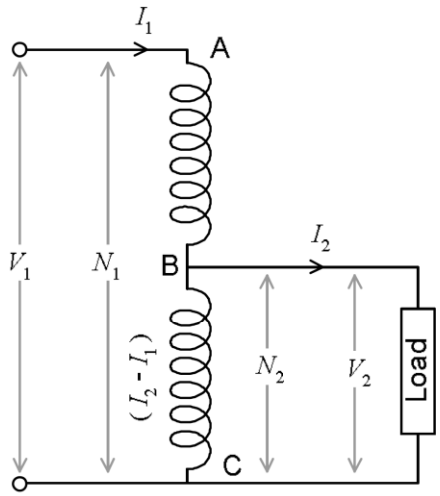
Advantages of Autotransformers

- **Lesser size and cost**
  - Size of the autotransformer will be lesser compared to two winding transformer of same capacity
  - There is saving in both copper and core material
- **Higher efficiency**
  - Autotransformer will be higher efficiency compared to two winding transformer of same capacity
  - This is because the effective winding resistance is less
- **Voltage regulation is better**
  - Lesser voltage drop for comparable load conditions
  - This is because the leakage reactance and also the resistance is less
- **Lower excitation current**
  - Since the core size is less, excitation current required will also be less

Limitations of Autotransformers

- **There is no electrical isolation between primary and secondary windings**
  - This is because some portions of the winding is shared between primary and secondary
- **Larger short circuit current**
  - This is because the effective winding impedance is less
- **Application limited to lower voltages only**
  - Since there is no electrical isolation between primary and secondary, higher voltages are not possible
- **Copper saving is considerable only when *k* is close to 1**
  - When the transformation ratio k is far different from unity, the copper saving is less

Transformed VA and Conducted VA (step down)



- Power is transferred from input to output by transformer action and conduction
- Volt amperes across winding section AB is transferred to section BC by transformer action

$$\text{Transformed VA} = V_{AB}\, I_{AB} = (V_1 - V_2) I_1$$

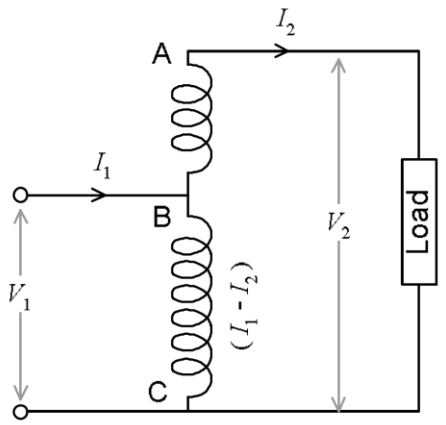
$$\text{Total input VA} = V_1 I_1$$

$$\text{Conducted VA} = \text{Total input VA} - \text{Transformed VA}$$

$$= V_1 I_1 - (V_1 - V_2) I_1 = V_2 I_1$$

$\text{VA rating} = V_1 I_1 = V_2 I_2$ $\quad$ $\text{Transformed VA} = (V_1 - V_2) I_1$ $\quad$ $\text{Conducted VA} = V_2 I_1$

Transformed VA and Conducted VA (step up)



- Power is transferred from input to output by transformer action and conduction
- Volt amperes across winding section AB receives power by transformer action

$$\text{Transformed VA} = V_{AB}\, I_{AB} = (V_2 - V_1) I_2$$

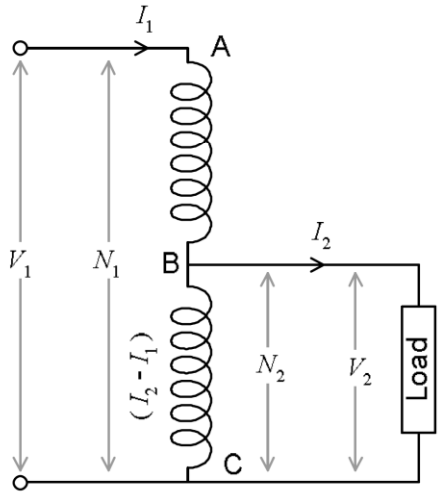
$$\text{Total output VA} = V_2 I_2$$

$$\text{Conducted VA} = \text{Total output VA} - \text{Transformed VA}$$

$$= V_2 I_2 - (V_2 - V_1) I_2 = V_1 I_2$$

$\text{VA rating} = V_1 I_1 = V_2 I_2$ $\quad$ $\text{Transformed VA} = (V_2 - V_1) I_2$ $\quad$ $\text{Conducted VA} = V_1 I_2$

Copper Saving in Autotransformer



Weight of conductor for winding section AB $\propto I_1(N_1 - N_2)$

Weight of conductor for winding section BC $\propto (I_2 - I_1)N_2$

Total weight of conductor $\propto I_1(N_1 - N_2) + (I_2 - I_1)N_2$

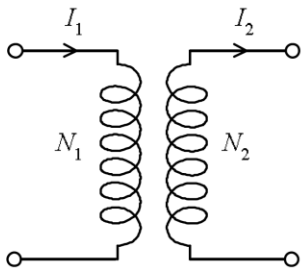
$\propto I_1N_1 - I_1N_2 + I_2N_2 - I_1N_2$

$\propto 2(I_1N_1 - I_1N_2)$

$\propto 2(N_1 - N_2)I_1$

$\because I_1N_1 = I_2N_2$

Copper Saving in Autotransformer



$$\text{Total weight of conductor for a 2-winding transformer} \propto I_1N_1 + I_2N_2$$

$$\propto 2I_1N_1$$

$$\frac{\text{Weight of conductor for autotransformer}}{\text{Weight of conductor for 2-winding transformer}} = \frac{2\left(N_1 - N_2\right)I_1}{2N_1I_1}$$

$$= 1 - \frac{N_2}{N_1} = 1 - k$$

$$k = \frac{N_2}{N_1}$$

$$\text{Saving copper for autotransformer} = k \times \text{Weight of conductor for 2-winding transformer}$$

Example 6.1

The primary and secondary voltages of an autotransformer are 500 V and 375 V secondary. If the current delivered to the load is 10 A. Find the primary current. Also find the copper saving compared to a two winding transformer of the same rating.

$$\text{Primary current } = \frac{375}{500} \times 10 = 7.5 \text{ A}$$

Example 6.1

The primary and secondary voltages of an autotransformer are 500 V and 375 V secondary. If the current delivered to the load is 10 A. Find the primary current. Also find the copper saving compared to a two winding transformer of the same rating.

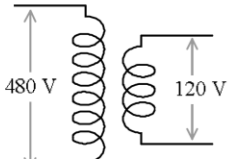
$$\text{Primary current} = \frac{375}{500} \times 10 = 7.5 \text{ A}$$

$$\text{Saving in copper} = k = \frac{375}{500} = 0.75 = 75\%$$

So copper to be used in the autotransformer is only 25% compared to what is required in two-winding transformer

Example 6.2

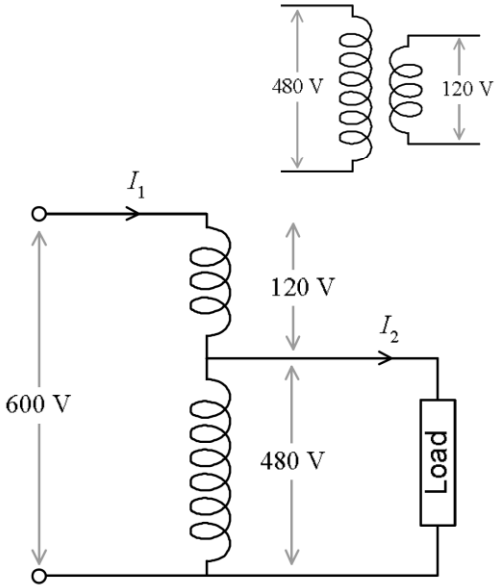
A 10 kVA, 480/120 V transformer connected as an autotransformer to produce 480 V with an input voltage of 600 V. Calculate the kVA rating, transformed power and conducted power of the autotransformer. If the efficiency at full load and UPF of the 2-winding transformer is 97.5 %, find the efficiency of the autotransformer at full load and 0.7 power factor.



$$\text{Current rating of the 120V winding} = \frac{10000}{120} = 86.33 \text{ A}$$

Example 6.2

A 10 kVA, 480/120 V transformer connected as an autotransformer to produce 480 V with an input voltage of 600 V. Calculate the kVA rating, transformed power and conducted power of the autotransformer. If the efficiency at full load and UPF of the 2-winding transformer is 97.5 %, find the efficiency of the autotransformer at full load and 0.7 power factor.



$$\text{Current rating of the 120V winding} = \frac{10000}{120} = 86.33 \text{ A}$$

$$\text{KVA rating of autotransformer} = V_1 I_1 = 600 \times 86.33$$

$$= 50 \text{ kVA}$$

$$\text{Transformed power} = (V_1 - V_2) I_1 = 120 \times 83.33 = 10 \text{ kVA}$$

$$\text{Conducted power} = V_2 I_1 = 480 \times 83.33 = 40 \text{ kVA}$$

Consider 2-winding operation at full and UPF; Efficiency is 97.5%

$$\text{Efficiency} = \frac{\text{Output}}{\text{Output+losses}} = \frac{10000 \times 1}{10000 \times 1 + \text{losses}} = 0.975 \quad \Rightarrow \quad \text{losses} = 256.4 \text{ W}$$

The same loss will be applicable for autotransformer at full load

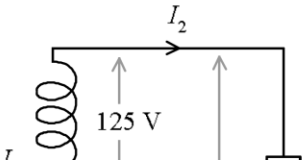
$$\text{Power output of the transformer} = \text{VA} \times \text{PF} = 50000 \times 0.7 = 35000 \text{ W}$$

$$\text{Power input} = \text{Output} + \text{losses} = 35000 + 256.4 = 35256.4 \text{ W}$$

$$\text{Efficiency} = \frac{\text{Output}}{\text{Input}} \times 100 = \frac{35000}{35256.4} \times 100 = 99.27\ \%$$

Example 6.3

A 5 kVA, 250/125 V transformer connected as an autotransformer to produce 375 V with an input voltage of 250V. Calculate the kVA rating, transformed power and conducted power of the autotransformer.

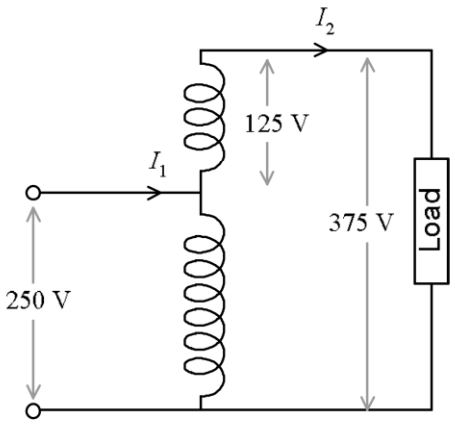


$$\text{Current rating of the 125V winding} = \frac{5000}{125} = 40 \text{ A}$$

$$\text{KVA rating } = V_2 I_2 = 375 \times 40 = 15 \text{ kVA}$$

Example 6.3

A 5 kVA, 250/125 V transformer connected as an autotransformer to produce 375 V with an input voltage of 250V. Calculate the kVA rating, transformed power and conducted power of the autotransformer.



$$\text{Current rating of the 125V winding} = \frac{5000}{125} = 40 \text{ A}$$

$$\text{KVA rating } = V_2 I_2 = 375 \times 40 = 15 \text{ kVA}$$

$$\text{Transformed power} = \left(V_2 - V_1\right) I_2 = 125 \times 40 = 5 \text{ kVA}$$

$$\text{Conducted power} = V_1 I_2 = 250 \times 40 = 10 \text{ kVA}$$

Three Phase Transformer

3-PHASE TRANSFORMER

Electric power is generated and transmitted using a three-phase system. In a three-phase circuit, the voltage is raised or lowered by means of three-phase transformers.

Three number of identical single-phase transformers can be suitably connected for use in a three-phase system and such a 3-phase transformer is called a bank of three phase transformer. Alternately, the three-phase transformer can be built on a single core.

CONSTRUCTION OF 3-PHASE TRANSFORMERS

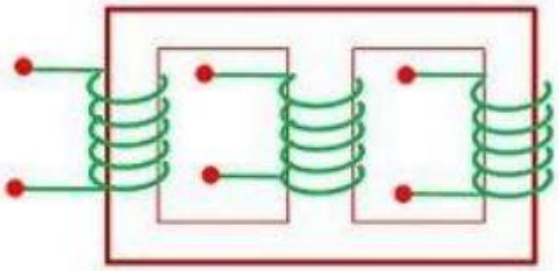
The three-phase transformer is constructed in two ways:

1. Three separate single-phase transformer is suitably connected for three-phase operation.
2. A single three-phase transformer in which the cores and windings for all the three phases are merged into a single structure.

Advantages of using a single 3-phase unit instead of a bank of 3 single-phase units are

- Single 'three-phase' unit occupies less space than a bank of three single-phase transformers
- Single 'three phase' unit is more economical
- The overall bus-bar structure, switchgear and installation of single 'three-phase' unit is simpler.

Core-type 3-phase transformer

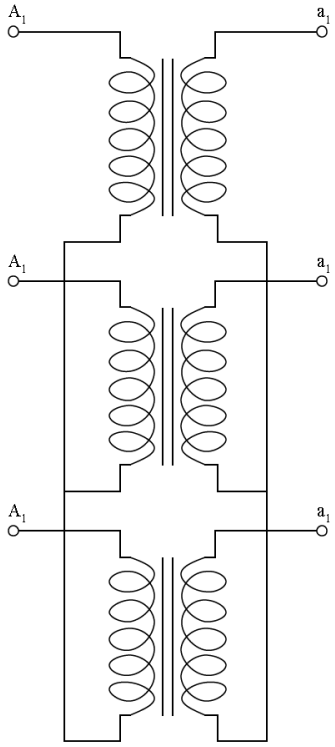


The core of the three phase transformer is usually made up of three limbs in the same plane. This can be built using stack lamination. The each leg of this core carries the low voltage and high voltage winding. The low voltage windings are insulated from the core than the high voltage windings. The low windings are placed next to the core with suitable insulation between the core and the low voltage windings. The high voltage windings are placed over the low voltage windings with suitable insulation between them.

Three Phase Transformers

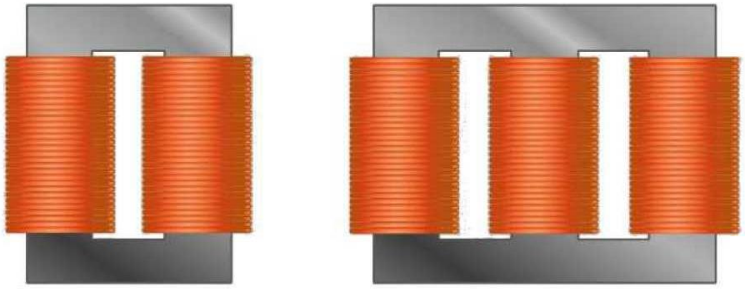


- Core type construction is normally adopted for three phase transformers
- Each limb has a primary and secondary winding on it
- The windings are connected in star or delta on both sides
- **A bank of three single phase transformers can also function as three phase transformer**



Three Phase Transformer – Core type and Shell type

Core type

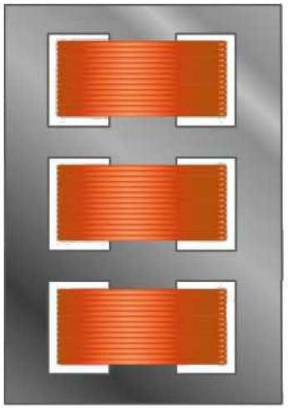


Single phase

Three phase

Shell type

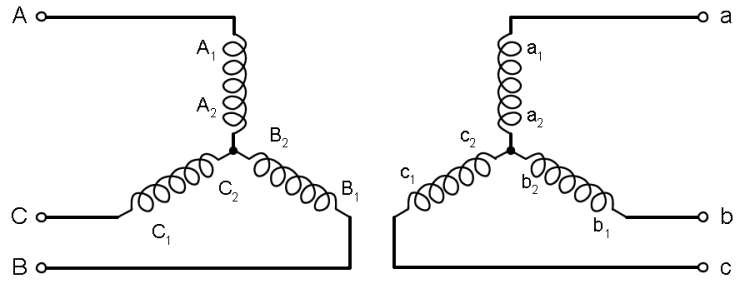




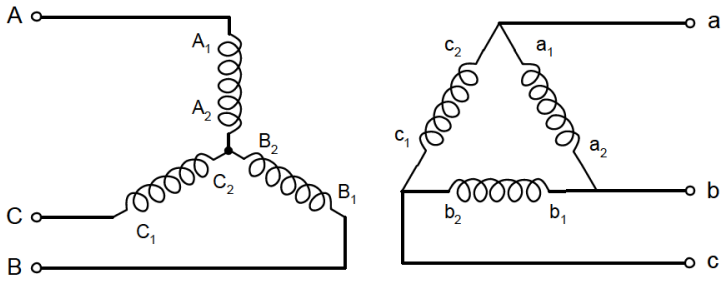
- In three phase core type construction, the windings are placed on the three limbs of the core
- In three phase shell type construction, a stacked construction is adopted as shown

Transformer Connections

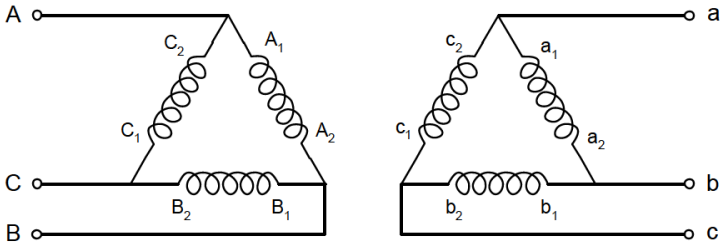
Star - Star



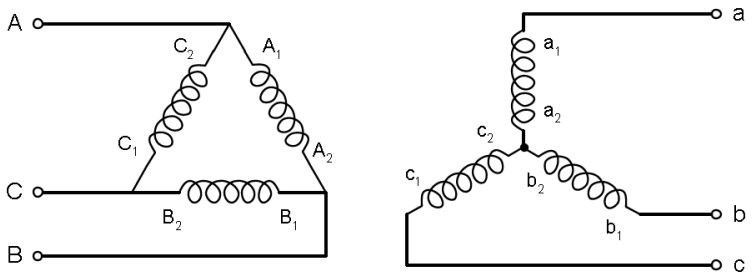
Star - Delta



Delta - Delta



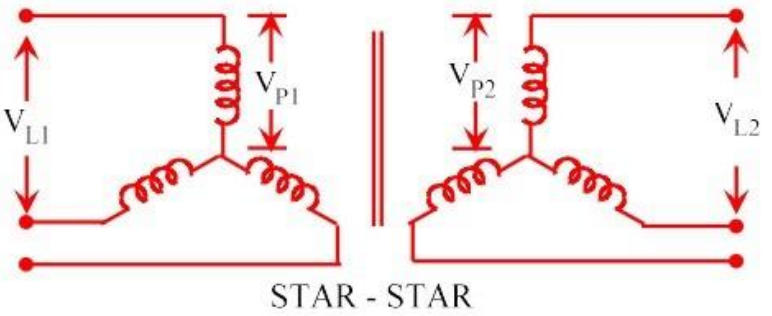
Delta - Star



3-PHASE TRANSFORMER CONNECTIONS

Windings of a three-phase transformer can be connected in various configurations as (i) star-star, (ii) delta-delta, (iii) star-delta, (iv) delta-star and (v) open delta. These configurations are explained below.

STAR-STAR CONNECTION



Transformation ratio, $K = \dfrac{\text{secondary phase voltage}}{\text{primary phase voltage}}$

$$V_{L1} = \sqrt{3} V_{P1}$$

$$V_{L2} = \sqrt{3} V_{P2}$$

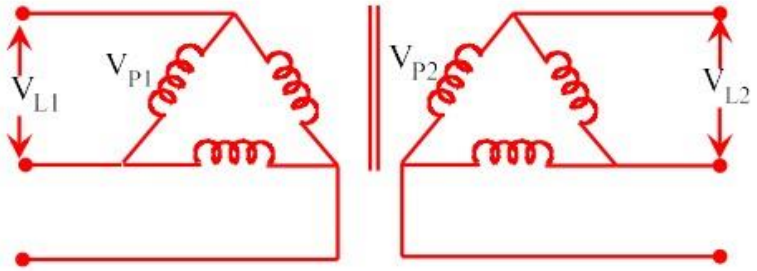
$$\frac{V_{L2}}{V_{L1}} = \frac{V_{P2}}{V_{P1}} = K$$

- Star-star connection is generally used for small, high-voltage transformers. Because of star connection, number of required turns/phase is reduced (as phase voltage in star connection is $1/\sqrt{3}$ times of line voltage only). Thus, the amount of insulation required is also reduced.
- The ratio of line voltages on the primary side and the secondary side is equal to the transformation ratio of the transformers.
- Line voltages on both sides are in phase with each other.
- This connection can be used only if the connected load is balanced.

3-PHASE TRANSFORMER CONNECTIONS

Windings of a three-phase transformer can be connected in various configurations as (i) star-star, (ii) delta-delta, (iii) star-delta, (iv) delta-star and (v) open delta. These configurations are explained below.

DELTA – DELTA CONNECTION



Transformation ratio, $K = \dfrac{\text{secondary phase voltage}}{\text{primary phase voltage}}$

$$V_{L1} = V_{P1}$$

$$V_{L2} = V_{P2}$$

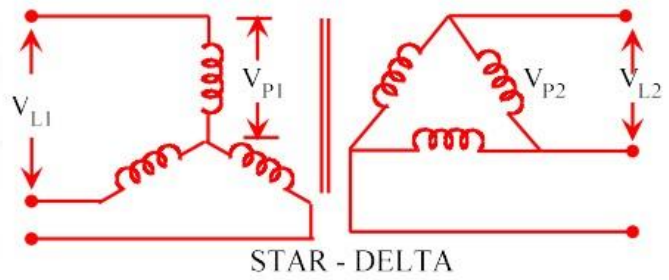
$$\frac{V_{L2}}{V_{L1}} = \frac{V_{P2}}{V_{P1}} = K$$

- This connection is generally used for large, low-voltage transformers. Number of required turns/phase is relatively greater than that for star-star connection.
- The ratio of line voltages on the primary and the secondary side is equal to the transformation ratio of the transformers.
- This connection can be used even for unbalanced loading.
- Another advantage of this type of connection is that even if one transformer is disabled, system can continue to operate in open delta connection but with reduced available capacity.

3-PHASE TRANSFORMER CONNECTIONS

Windings of a three-phase transformer can be connected in various configurations as (i) star-star, (ii) delta-delta, (iii) star-delta, (iv) delta-star and (v) open delta. These configurations are explained below.

STAR-DELTA CONNECTION



STAR - DELTA

Transformation ratio, $K = \dfrac{\text{secondary phase voltage}}{\text{primary phase voltage}}$

$$V_{L1} = \sqrt{3} V_{P1}$$

$$V_{L2} = V_{P2}$$

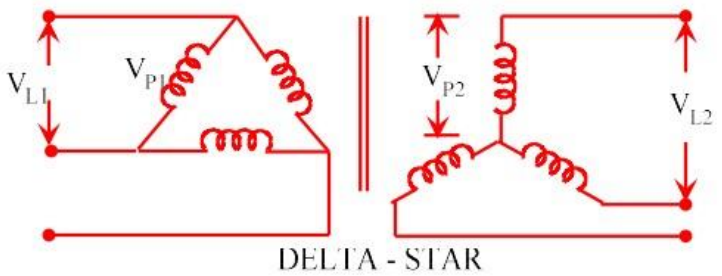
$$\frac{V_{L2}}{V_{L1}} = \frac{1}{\sqrt{3}} \frac{V_{P2}}{V_{P1}} = \frac{K}{\sqrt{3}}$$

- This connection is mainly used in step down transformer at the substation end of the transmission line.
- The ratio of secondary to primary line voltage is $1/\sqrt{3}$ times the transformation ratio.
- There is 30° shift between the primary and secondary line voltages.

3-PHASE TRANSFORMER CONNECTIONS

Windings of a three-phase transformer can be connected in various configurations as (i) star-star, (ii) delta-delta, (iii) star-delta, (iv) delta-star and (v) open delta. These configurations are explained below.

DELTA – STAR CONNECTION



DELTA - STAR

Transformation ratio, $K = \dfrac{\text{secondary phase voltage}}{\text{primary phase voltage}}$

$$V_{L1} = V_{P1}$$

$$V_{L2} = \sqrt{3} V_{P2}$$

$$\frac{V_{L2}}{V_{L1}} = \frac{\sqrt{3} V_{P2}}{V_{P1}} = \sqrt{3} K$$

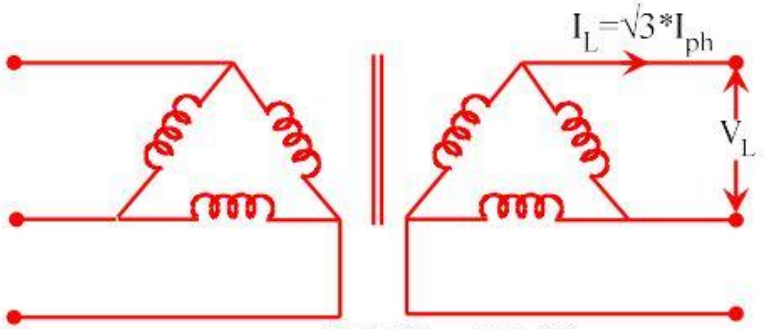
- This type of connection is mainly used in step-up transformer at the beginning of transmission line.
- The ratio of secondary to primary line voltage is √3 times the transformation ratio.
- There is 30° shift between the primary and secondary line voltages.

OPEN DELTA (V – V) CONNECTION

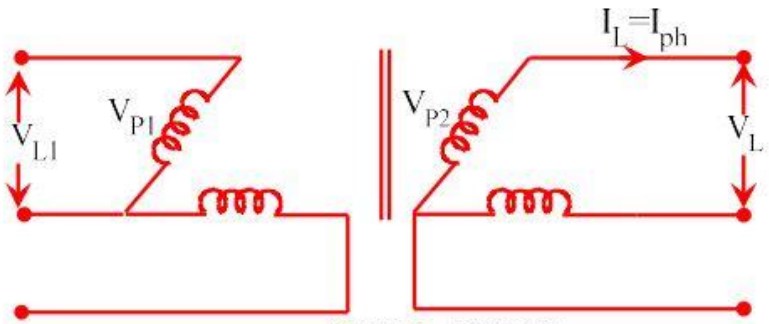
An open delta connection transformer uses two single-phase transformers to provide a three-phase supply to the load. An open delta connection system is also called a V-V system.

When one phase of the three single phase transformers connected in delta-delta bank is unable to operate or damaged. Then supply to the load can be continued with the remaining two phase of transformers in open delta connection. Open delta connection systems are usually used only in emergency conditions, as their efficiency is low when compared to delta-delta (closed delta) systems (which are used during standard operations).

The load which can be delivered by a V – V bank is not 2/3 of the capacity of Δ - Δ bank, but only 57.7% of it.



DELTA - DELTA



OPEN - DELTA

When it connected in delta-delta so VA rating is given by,

$$S_{\Delta\Delta} = \sqrt{3} V_L I_L = 3 V_L I_{ph}$$

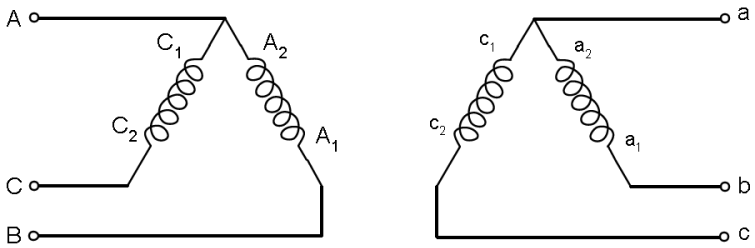
When it connected in open-delta so VA rating is given by,

$$S_{VV} = \sqrt{3} V_L I_L = \sqrt{3} V_L I_{ph}$$

Hence, $$\frac{S_{VV}}{S_{\Delta\Delta}} = \frac{1}{\sqrt{3}} = 0.577$$

For example, if three transformers of 10kVA rating are connected to form a Δ - Δ bank, the rating of the bank is 3 x 10 = 30kVA. But if one of the transformer is removed and the operation is carried out with the remaining to transformers by V – V connection, the rating now will be 30 x 0.577 = 17.3kVA only and not 20kVA.

V-V Connection (Open Delta)



- If one transformer in a $\Delta - \Delta$ bank is removed, then it becomes a V – V connection
- It also points to the fact if one transformer in a three phase bank goes non-functional, the other two can still provide three phase supply with reduced rating

Reduction in power rating

$$\Delta - \Delta \text{ Capacity} = \sqrt{3}\, V_L I_L = \sqrt{3}\, V_L \sqrt{3}\, I_{Ph}$$
$$= 3\, V_L\, I_{Ph}$$

$$\text{V - V Capacity} = \sqrt{3}\, V_L I_L = \sqrt{3}\, V_L\, I_{Ph}$$

$$\frac{\text{V} - \text{V Capacity}}{\Delta - \Delta \text{ Capacity}} = \frac{\sqrt{3}\, V_L\, I_{Ph}}{3\, V_L\, I_{Ph}} = 0.577$$

A V-V connected transformer will have 57.7% capacity of a corresponding Δ–Δ transformer

Example 5.1

A 11 kV/415 V Δ-Y transformer has a line current of 100 A in the LV side. Find the line current in the HV side. Also find the turns ratio. Neglect magnetising current.

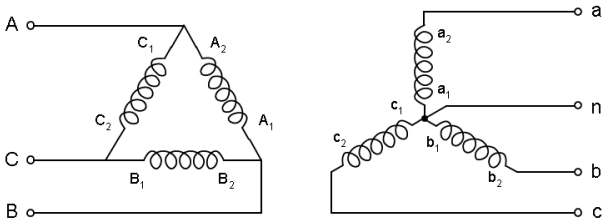
HV side phase voltage = 11000 V

$$\text{LV side phase voltage} = \frac{415}{\sqrt{3}} = 239.6 \text{ V}$$

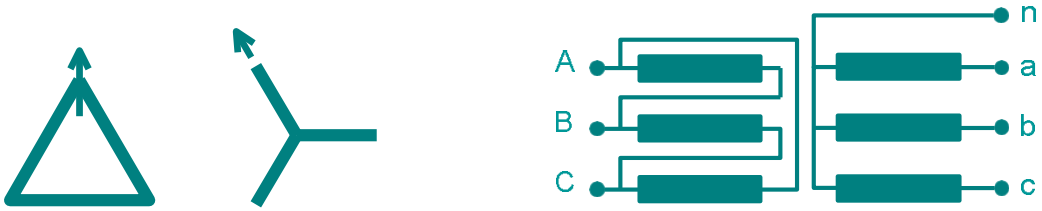
$$\text{Turns ratio} = \frac{N_1}{N_2} = \frac{11000}{239.6} = 45.9$$

$$\text{HV side phase current} = \frac{100}{45.9} = 2.18 \text{ A}$$

$$\text{HV side line current} = \sqrt{3} \times 2.18 = 3.77 \text{ A}$$

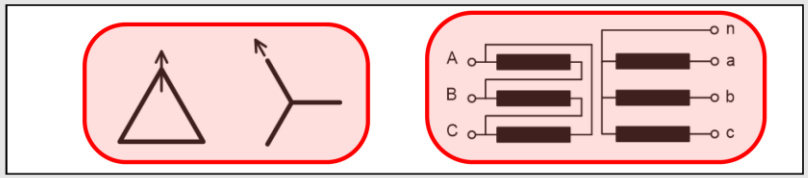


Transformer Vector Groups



Typical Transformer Name Plate

3	PHASE TRANSFORMER		Sl. No		123456789
KVA		63	FREQUENCY	Hz	50
STANDARD		IS : 2026/1180	COOLING		ONAM
VOLTS AT NO LOAD	HV	11000	VECTOR GROUP		Dyn11
	LV	433	CORE & WDG	kg	240
AMPERES	HV	3.31	MASS OF OIL	kg	140
	LV	84.0	TOTAL MASS	kg	505
IMPEDANCE	%	4.5	TEMP RISE IN OIL		35°C
LOAD LOSSES (W)		1350	TEMP RISE IN WDG		40°C
NO LOAD LOSSES (W)		225	YEAR OF MFG		2018



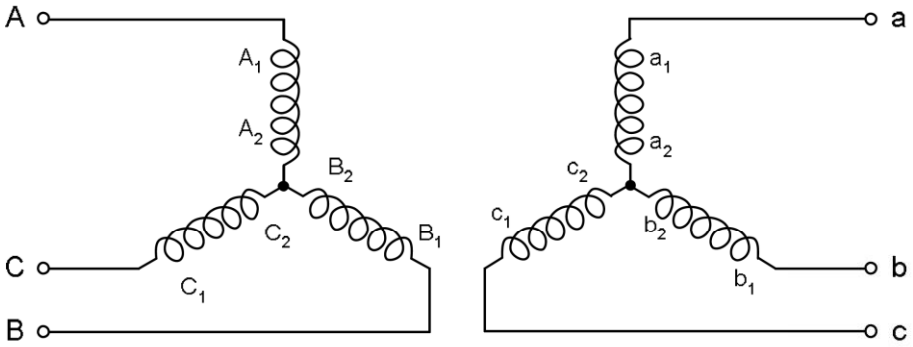
The **vector group** designation indicates the windings configurations and the difference in phase angle between them

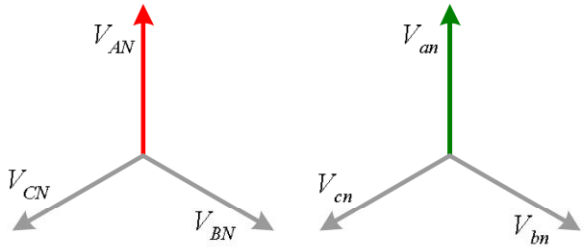
D – specifies primary is delta connected
y – specifies secondary star connected
n – indicates neutral brought out
11 – specifies the phase angle is 30° leading

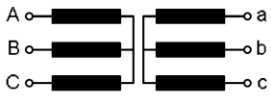


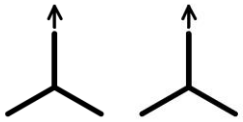
Yy0 Vector Group

Star – star connection with zero phase shift





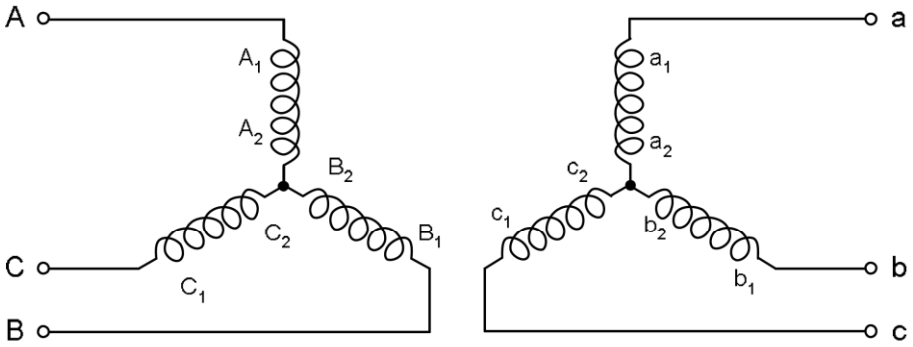


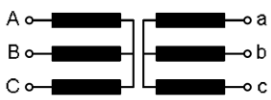




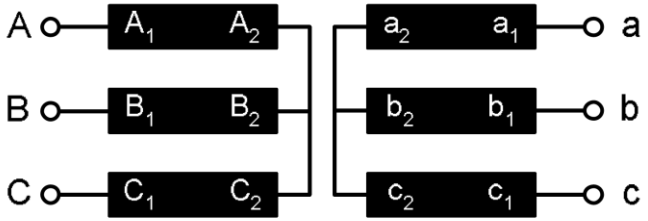
Yy0 Vector Group

Star – star connection with zero phase shift



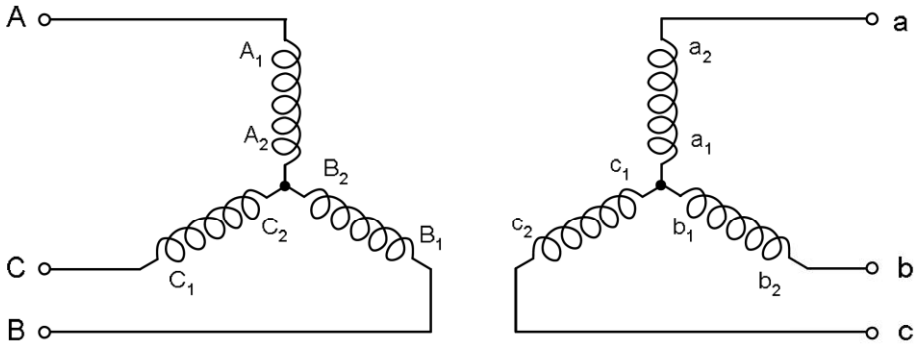


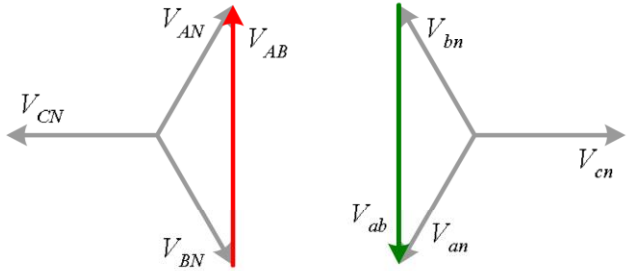


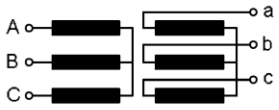


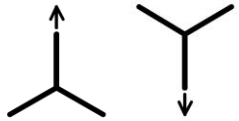
Yy6 Vector Group

Star – star connection with phase shift of 180°





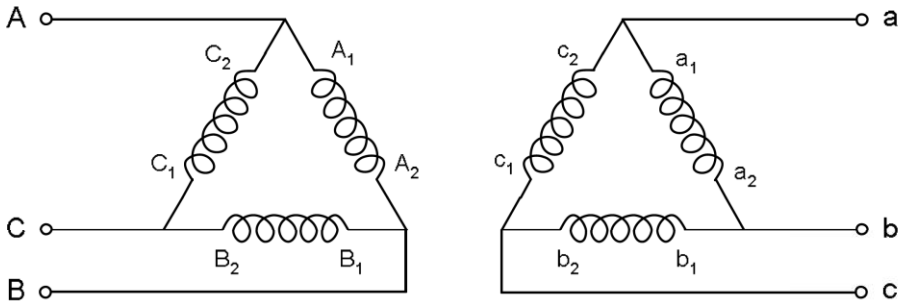


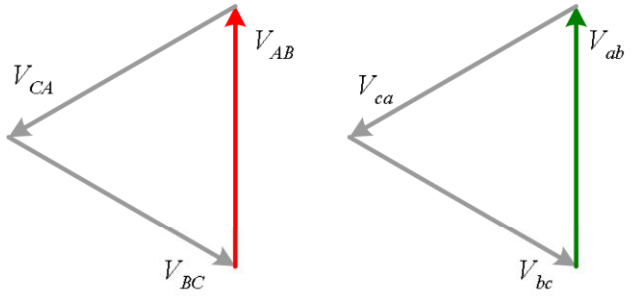


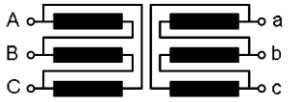


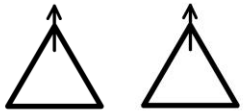
Dd0 Vector Group

Delta – delta connection with zero phase shift





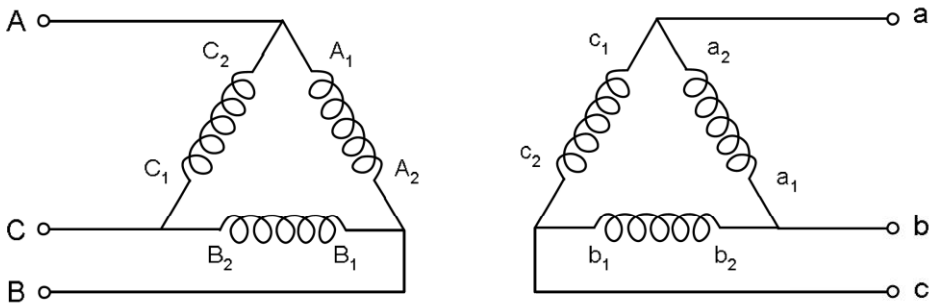


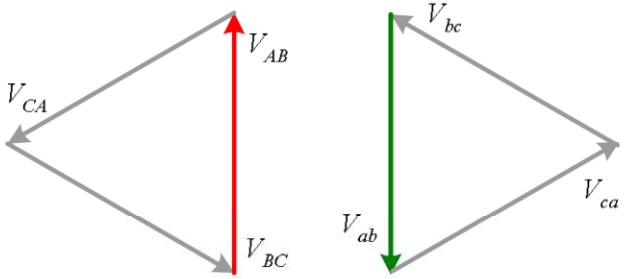


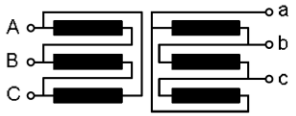


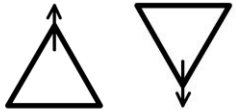
Dd6 Vector Group

Delta – delta connection with phase shift of 180°





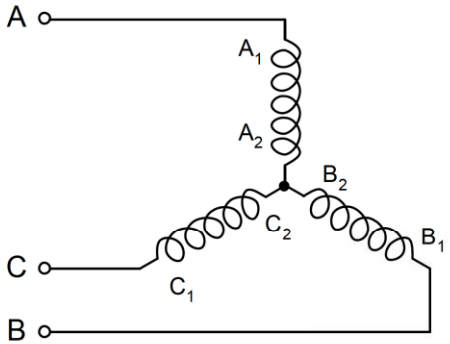


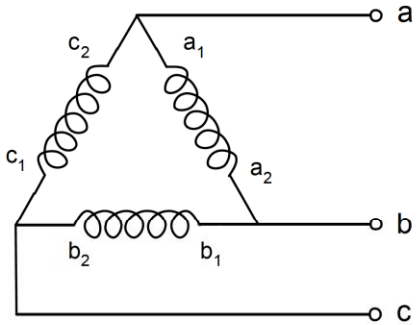


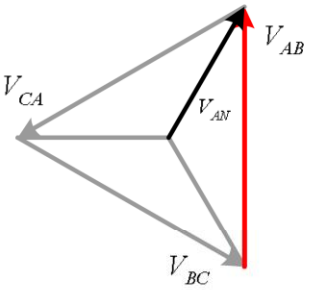


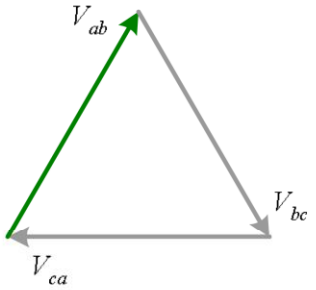
Yd1 Vector Group

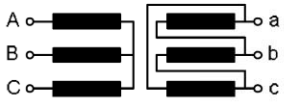
Star – delta connection with phase shift of 30° lag

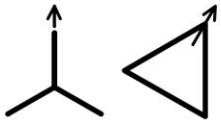








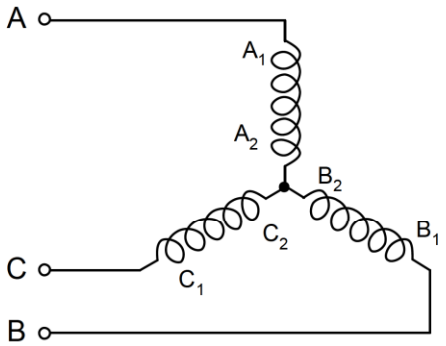


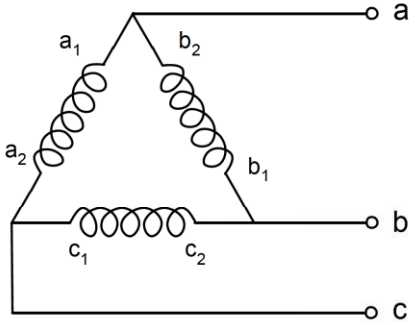


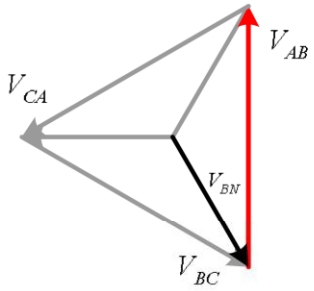


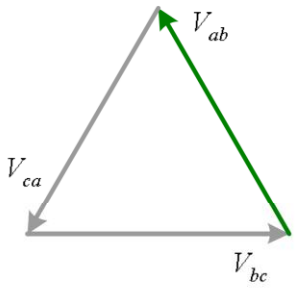
Yd11 Vector Group

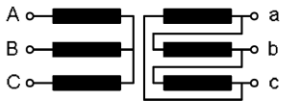
Star – delta connection with phase shift of 30° lead

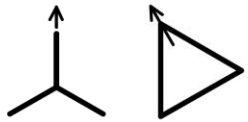








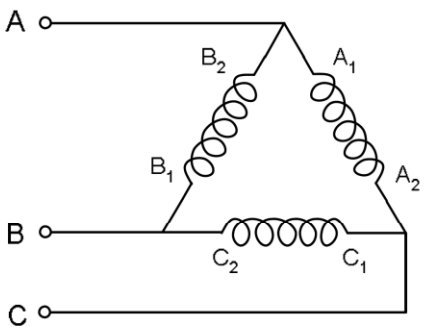


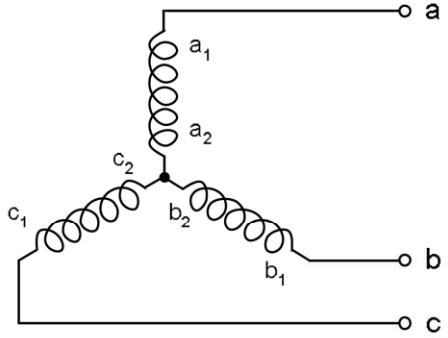


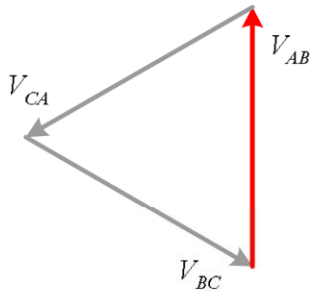


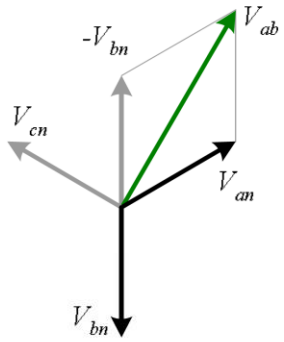
Dy1 Vector Group

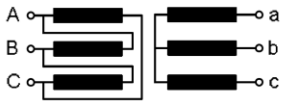
Delta – star connection with phase shift of 30° lag

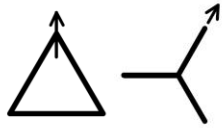








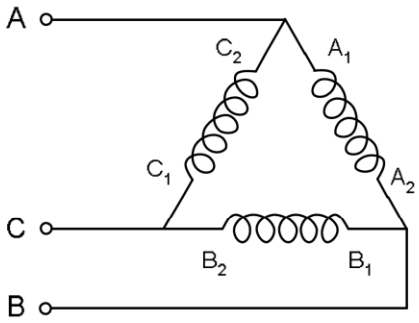


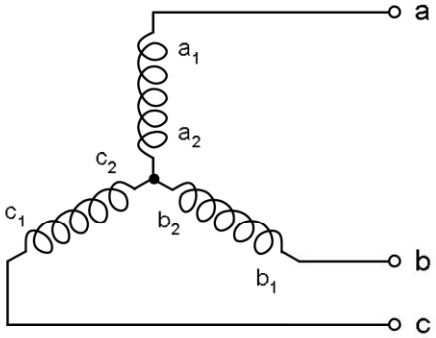


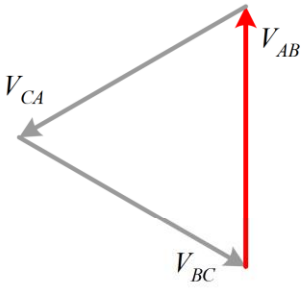


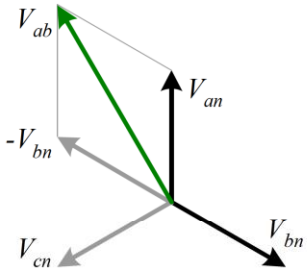
Dy11 Vector Group

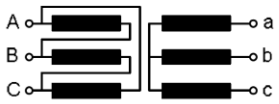
Delta – star connection with phase shift of 30° lead

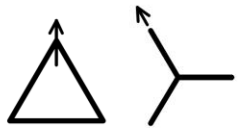






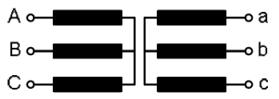
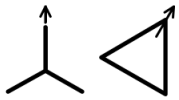
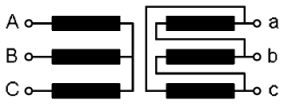
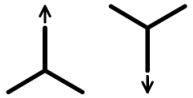
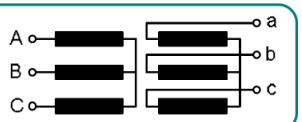
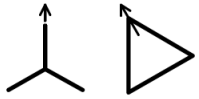
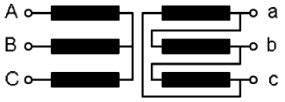


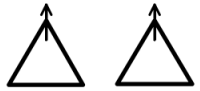
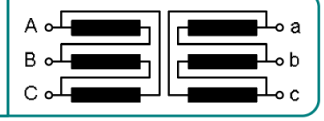
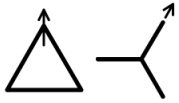
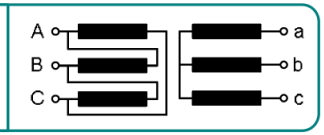
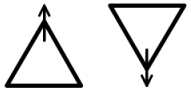
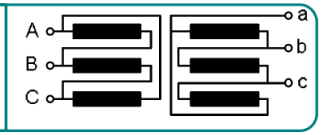
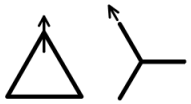
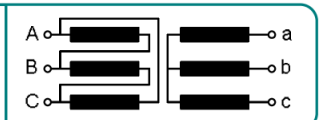






Vector Groups of Three Phase Transformers

Phasor Symbol	Displacement Diagram	Winding Connections
Yy0		
Yd1		
Yy6		
Yd11		

Phasor Symbol	Displacement Diagram	Winding Connections
Dd0		
Dy1		
Dd6		
Dy11		

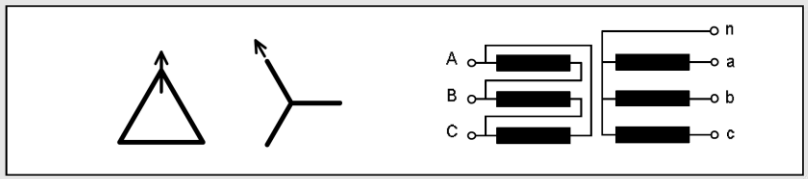
Group	O'clock	Phase Difference	Connection
Group 1	0 o'clock	0°	Y-Y, Δ-Δ
Group 2	6 o'clock	180°	Y-Y, Δ-Δ
Group 3	1 o'clock	-30°	Y-Δ, Δ- Y
Group 4	11 o'clock	+30°	Y-Δ, Δ- Y

Note: The only transformer in the same group may be connected in parallel.

For example, a star-star, 3-phase transformers can be parallel with another 3-phase transformer whose windings are either connected in Y-Y or Δ-Δ. The Δ-Δ transformer cannot be parallel with Y-Δ transformer.

Dyn11 Connection

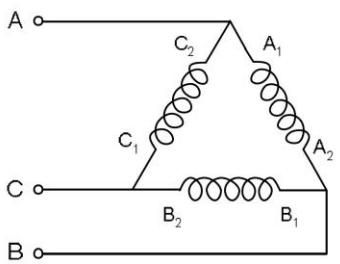
3	PHASE TRANSFORMER		Sl. No		123456789
KVA		63	FREQUENCY	Hz	50
STANDARD		IS : 2026/1180	COOLING		ONAM
VOLTS AT NO LOAD	HV	11000	VECTOR GROUP		Dyn11
	LV	433	CORE & WDG	kg	240
AMPERES	HV	3.31	MASS OF OIL	kg	140
	LV	84.0	TOTAL MASS	kg	505
IMPEDANCE	%	4.5	TEMP RISE IN OIL		35°C
LOAD LOSSES (W)		1350	TEMP RISE IN WDG		40°C
NO LOAD LOSSES (W)		225	YEAR OF MFG		2018

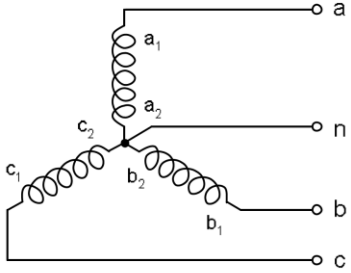


Dyn11 vs Dy11

In Dyn11, **neutral terminal** is taken out in the secondary

Phase shift is 30° lead





Turn per phase

Example 2.3. A 3- phase step-down transformer is connected to 6.6 kv mains and takes 10 A. Calculate the secondary line voltage, line current and output for the following connections:

(i) Δ/Δ (ii) Y/Y (iii) Δ/Y and (iv) Y/Δ

The ratio of turns per phase is 12. Neglect losses.

Solution : Primary line voltage, $V_{L1} = 6.6 \text{ kv} = 6{,}600 \text{ V}$

Primary line current, $I_{L1} = 10 \text{ A}$

$$\text{Turn-ratio} = \frac{N_1}{N_2} = 12$$

(i) Δ–Δ Connections

Primary phase voltage, $V_{P1} = V_{L1} = 6{,}600 \text{ V}$ ∵ winding is delta-connected

Secondary line voltage, $V_{L2} = V_{P2} = V_{P1} \times \frac{N_2}{N_1} = \frac{6{,}600}{12} = 550 \text{ V}$ Ans.

Primary phase current, $I_{P1} = \frac{I_{L1}}{\sqrt{3}} = \frac{10}{\sqrt{3}} = 5.773 \text{ A}$

Secondary line current, $I_{L2} = \sqrt{3}\, I_{P2} = \sqrt{3} \times I_{P1} \times \frac{N_1}{N_2} = \sqrt{3} \times 5.773 \times 12 = 120 \text{ A}$ Ans.

(ii) Y/Y Connections

$$\text{Primary phase voltage, } V_{P1} = \frac{V_{L1}}{\sqrt{3}} = \frac{6{,}600}{\sqrt{3}} \qquad \because \text{winding is star-connected}$$

$$\text{Secondary line voltage, } V_{L2} = \sqrt{3}\ V_{P2} = \sqrt{3}\ V_{P1} \times \frac{N_2}{N_1} = \sqrt{3} \times \frac{6{,}600}{\sqrt{3}} \times \frac{1}{12} = 550 \text{ V Ans.}$$

$$\text{Primary phase current, } I_{P1} = I_{L1} = 10 \text{ A}$$

$$\text{Secondary line current, } I_{L2} = I_{P2} = I_{P1} \frac{N_1}{N_2} = 10 \times 12 = 120 \text{ A Ans.}$$

(iii) Δ/Y Connections

$$\text{Primary phase voltage, } V_{P1} = V_{L1} = 6{,}600 \text{ V}$$

$$\text{Secondary line voltage, } V_{L2} = \sqrt{3}\ V_{P2} = \sqrt{3} \times V_{P1} \times \frac{N_1}{N_2} = \sqrt{3} \times \frac{6{,}600}{12} = 952.6 \text{ V Ans.}$$

$$\text{Primary phase current, } I_{P1} = \frac{I_{L1}}{\sqrt{3}} = \frac{10}{\sqrt{3}} \text{ A}$$

$$\text{Secondary line current, } I_{L2} = I_{P2} = I_{P1} \times \frac{N_1}{N_2} = \frac{10}{\sqrt{3}} \times 12 = 69.28 \text{ A Ans.}$$

(iv) Y/Δ Connections

$$\text{Primary phase voltage, } V_{P1} = \frac{V_{L1}}{\sqrt{3}} = \frac{6{,}600}{\sqrt{3}} \text{ V}$$

$$\text{Secondary line voltage, } V_{L2} = V_{P2} = V_{P1} \times \frac{N_2}{N_1} = \frac{6{,}600}{\sqrt{3}} \times \frac{1}{12} = 317.5 \text{ V Ans.}$$